

COURSE: AGENTIC AI & LARGE LANGUAGE MODELS

Vector Databases

From Fundamentals to Advanced Concepts

B.TECH FINAL YEAR

COMPUTER ENGINEERING

PART 1 OF 4



This Lecture Covers



Vectors & Embeddings



Semantic vs Keyword Search



Vector DB vs SQL Database



AI Applications & Use Cases



Powering **ChatGPT** · **RAG** · **AI Agents** ·
Recommendations

Learning Outcomes

After completing this lecture, students will be able to:

1

Explain the Need

Why Vector Databases are needed in modern AI

2

Differentiate Databases

SQL vs Vector DB — architecture & use cases

3

Understand Vectors

Vectors as numerical representations of data

1

Explain Semantic Search

Meaning-based vs keyword-based retrieval

2

Vector DB Architecture

Indexing and similarity search mechanisms

3

LLM & RAG Integration

How Vector DBs power ChatGPT and RAG systems

MOTIVATION

Why Modern AI Needs Vector Databases

Traditional Search

Query: "Laptop for AI"

→ Returns **ONLY** exact keyword matches

✗ Misses: GPU Laptop, AI Workstation, Deep Learning PC

AI Semantic Search

Query: "Best computer for machine learning"

→ Also returns: AI Laptop · GPU Laptop · Deep Learning Workstation · Data Science Notebook

✓ Understands meaning, not just words



Key Idea: AI understands meaning, not just keywords.

Powered by Vector Search



ChatGPT



Microsoft Copilot



Amazon Product Search



Google Gemini



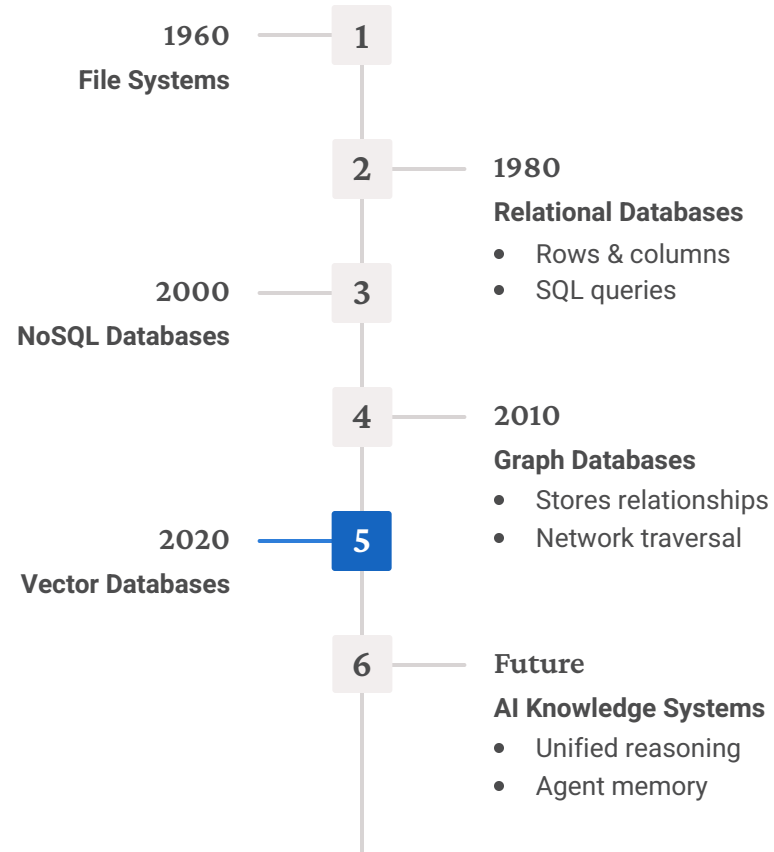
Netflix Recommendations

Evolution of Databases

- Files only
- No querying

- JSON & documents
- Flexible schema

- Semantic meaning
- AI-powered search



Each generation solved the limitations of the previous one.

Database Generations at a Glance

Database	Best For	Example Technology
File System	Files & Documents	FAT, NTFS
SQL	Structured Data	MySQL, PostgreSQL
NoSQL	JSON Documents	MongoDB, Redis
Graph	Relationships	Neo4j, ArangoDB
Vector DB	Semantic AI Search	Pinecone, Milvus

  **Vector Databases are the foundation of all modern AI search systems.**

  **Traditional SQL cannot perform semantic or similarity-based search.**

Types of Data

Structured Data

- Organized in rows & columns
- Easy to search with SQL
- Example: Employee Table (ID, Name, Salary)

Semi-Structured Data

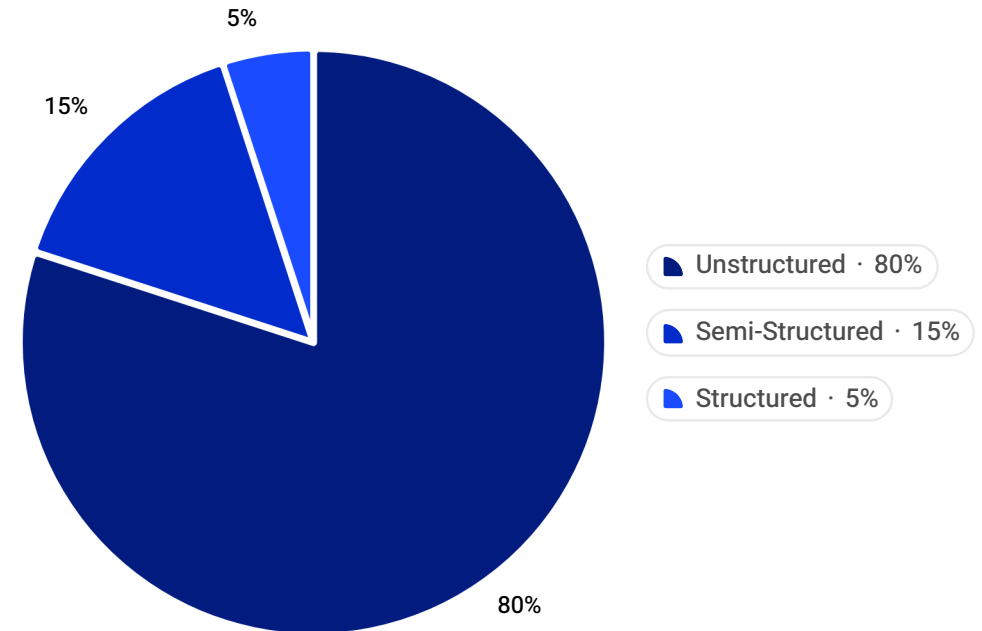
- Has some organization but no fixed schema
- Examples: JSON, XML, Emails, Logs

Unstructured Data

- No predefined format
- Examples: PDFs, Images, Videos, Audio
- Social Media, Medical Reports, Research Papers

⚠️ **Challenge:** Over **80%** of enterprise data is unstructured. Traditional SQL cannot efficiently search it.

Enterprise Data Distribution



Source: Industry estimates — IDC, Gartner



Hospitals

Medical reports, scans



Banks

Transaction notes, contracts

What is a Vector?

 **Definition:** A vector is an ordered list of numbers representing an object in multidimensional space.

Dog

[0.24, 0.82, -0.41, ... 768 dimensions]

Human

[0.61, -0.73, 0.12, ... 1536 dimensions]

 **Why?** Instead of storing words, AI stores **numerical meaning**.

Key Concepts in Vector Space



Dimension

Number of values in a vector. Higher = more nuance.



Magnitude

The length/size of the vector in space.



Direction

Orientation encodes semantic meaning.



Distance & Similarity

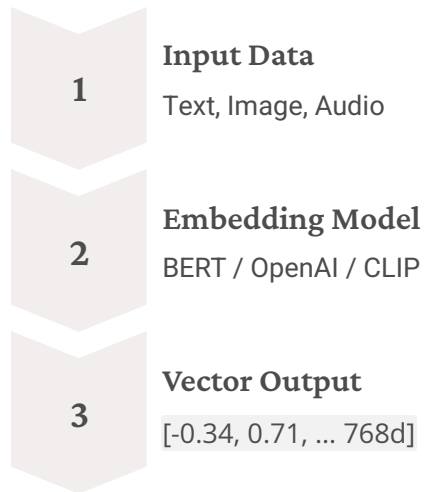
Closeness of two vectors = semantic similarity.

What are Embeddings?



Definition: Embeddings are numerical representations of data generated by deep learning models.

How Embeddings Are Generated



Popular Embedding Models

01

Word2Vec & FastText

Word-level models, Google & Meta.

02

BERT

Contextual transformer embeddings.

03

Sentence Transformers

Sentence-level similarity tasks.

04

OpenAI & Gemini

State-of-the-art commercial APIs.

Keyword Search vs Semantic Search

Keyword Search

Query: "AI Books"

→ Returns ONLY documents containing "AI Books"

✗ Misses related topics entirely

Semantic Search

Query: "Artificial Intelligence Learning Resources"

→ Also returns: Machine Learning Books · Deep Learning · LLM Books · Neural Networks

✓ Understands context and meaning

Feature Comparison

Feature	Keyword Search	Semantic Search
Matching	Exact Match	Meaning-Based
Technology	SQL / Inverted Index	Vector Database
Speed	Very Fast	Intelligent
Context Awareness	None	Context-Aware



💡 **Semantic Search** powers ChatGPT, Google, and all modern AI assistants.

Traditional Database vs Vector Database

Side-by-Side Comparison

Feature	SQL Database	Vector Database
Data Storage	Rows & Columns	High-Dimensional Vectors
Search Type	Keyword / Exact Match	Semantic Similarity Search
Data Type	Structured Data	Unstructured Data
Query Language	SQL	Vector Query (ANN)
Use Case	Banking, ERP, Payroll	ChatGPT, RAG, AI Agents

Applications

SQL Database

- Banking & Finance
- ERP Systems
- Payroll Management
- Inventory Control

Vector Database

- ChatGPT & LLMs
- AI-Powered Search
- Recommendation Systems
- RAG Pipelines
- AI Agents & Memory

Recap & What's Next

Key Takeaways

- ✓ Modern AI Understands Meaning
- ✓ Data is Converted into Vectors
- ✓ Embeddings Capture Semantic Meaning
- ✓ Vector DBs Store Embeddings
- ✓ Similarity Search Replaces Keywords

Coming Up in Part 2

Vector DB Architecture

HNSW, IVF, Flat Index

RAG Pipelines

LLMs + Vector DB integration

Popular Vector DBs

Pinecone, Weaviate, Chroma

Agentic AI

Vector DB as agent memory

  **Next:** Vector Database Architecture & RAG Integration

Coming Up: Part 2

Vector Database Architecture & Real-World AI Systems



Vector DB Architecture

- HNSW, IVF, and Flat Index strategies
- How similarity search works at scale



RAG Pipelines

- Retrieval-Augmented Generation
- How LLMs like GPT-4 & Gemini use Vector DBs



Popular Vector Databases

- Pinecone · Weaviate · Chroma
- Qdrant · Milvus — features & trade-offs



Agentic AI Integration

- AI agents using Vector DBs as long-term memory
- Knowledge retrieval for autonomous systems



Course: Agentic AI & Large Language Models — Department of Computer Engineering, B.Tech Final Year